A Continuous Hidden Markov Model for Synthetic Equity Returns: Emission-Family Ablation and Rank-Reordering Copula Extensions

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Abstract

We train a continuous hidden Markov model (CHMM) by Baum-Welch / Expectation-Conditional-Maximization on raw annualized excess log returns and show that, at a single moderate state resolution ($K = 18$), it reproduces all three canonical stylized facts of equity returns (heavy tails, negligible linear autocorrelation, persistent volatility clustering) without the Poisson jump mechanism required by the recent discrete-HMM baseline of Alswaidan et al. Three emission families are compared under identical training: Gaussian (CHMM-N), Student-t with per-state degrees of freedom (CHMM-t), and Laplace (CHMM-L). On 2014-2024 SPY in-sample (T = 2,516) and 2024-2026 out-of-sample (T = 572) data, all three CHMM variants reach KS pass rates above 94% in-sample and 80% out-of-sample, versus 0% for both discrete-HMM variants of the prior paper, 23% for GARCH(1,1), and 99.6% for an i.i.d. bootstrap ceiling. The rank-reordering copula extension composes cleanly on top of the per-asset CHMM marginals: across six tickers the Student-t copula (profile-MLE $\nu^* = 6$) reduces off-diagonal correlation MAE to 0.027 against 0.076 for a Single Index Model baseline, while preserving every per-asset marginal exactly. The framework separates cleanly into *Pipeline A* (per-ticker independent CHMM) and *Pipeline B* (cross-asset dependence on Pipeline A marginals), reported through self-contained tables and figures.

CCS Concepts

• **Computing methodologies** → **Unsupervised learning**; • **Theory of computation** → *Dynamic programming*; • **Applied computing** → Economics.

Keywords

continuous hidden Markov model, Baum-Welch, stylized facts, volatility clustering, Student-t copula, Single Index Model, synthetic financial data

1 Introduction

Synthetic financial time series are increasingly used for back-testing, stress testing, and training of downstream risk and trading models. A generator is useful for these tasks only if it reproduces the three canonical stylized facts of equity returns: heavy-tailed marginals, near-zero linear autocorrelation of returns, and persistent positive autocorrelation of absolute returns. Classical Gaussian hidden Markov models are widely believed to fail on the third fact because, as Rydén et al. [9] showed at state counts $K \in \{2, 3\}$, the geometric sojourn-time distribution decays too fast to sustain volatility clustering.

The recent discrete-HMM framework of Alswaidan et al. [1] works around this limitation by *quantizing* returns into Laplace quantile bins, frequency-counting transitions between bins, and adding a Poisson jump mechanism to reach observed tail heaviness. The quantization makes Kolmogorov-Smirnov (KS) tests fail by construction on the simulated return series; the jump mechanism introduces two free hyperparameters (ϵ for jump probability, λ for mean jump duration).

This paper shows that both workarounds are unnecessary. A continuous HMM trained by Expectation-Maximization (EM) on raw returns at a single moderate state count ($K = 18$, selected by a seven-metric sweep) matches all three stylized facts, passes KS / Anderson-Darling tests uniformly, and eliminates both hyperparameters of the prior discrete baseline. Three emission families (Gaussian, Student-t with per-state ν, Laplace) are compared under identical training; the Student-t and Laplace variants close the residual kurtosis gap of the Gaussian variant without sacrificing autocorrelation fidelity.

To extend the framework from single-asset to multi-asset synthesis we introduce the distinction between *Pipeline A* (per-ticker independent CHMM fits, no cross-asset coupling) and *Pipeline B* (SIM [10] and Gaussian / Student-t copula [3, 11] constructions on top of Pipeline A marginals). Pipeline A answers the univariate question (does the CHMM fit each ticker); Pipeline B answers the joint question (how well do the marginals compose).

2 Methods

2.1 CHMM Definition and EM Updates

Denote the annualized daily excess log-return series by $G_{1:T}$. A CHMM with K hidden states has parameters $\theta = \{\pi, T, \{\phi_k\}\}$: initial state distribution π, $K \times K$ transition matrix T with $T_{ij} = P(s_{t+1} = j \mid s_t = i)$, and per-state emission distributions ϕ_k. Three emission families are considered:

$$\text{CHMM-N}: \quad G_t \mid s_t = k \sim N(\mu_k, \sigma_k^2), \tag{1}$$

$$\text{CHMM-t}: \quad G_t \mid s_t = k \sim t_{\nu_k}(\mu_k, \sigma_k), \tag{2}$$

$$\text{CHMM-L}: \quad G_t \mid s_t = k \sim \text{Laplace}(\mu_k, b_k). \tag{3}$$

Training is by standard log-space forward-backward recursions. The M-step is closed-form Baum-Welch for CHMM-N; Expectation-Conditional Maximization (ECM) with a golden-section search over $\nu_k \in [2.5, 100]$ for CHMM-t [6, 8]; and weighted median and weighted mean-absolute-deviation estimators for CHMM-L. All three families use a quantile-based initialization that places state means on equi-spaced quantiles of $G_{1:T}$. Convergence tolerance is 10^{-4} on the log-likelihood; default iteration count is 60.

2.2 Cross-Asset Dependence (Pipeline B)

Let $\mathbf{R}_{\text{IS}} \in \mathbb{R}^{T \times d}$ be the d-asset return matrix and let the market-factor column be indexed by m. The SIM construction is

$$G_{j,t} = \alpha_j + \beta_j G_{m,t} + \eta_{j,t}, \tag{4}$$

fitted by OLS per non-market asset $j \neq m$, with simulation via $\hat{G}^{(p)}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G^{(p)}_{m,t} + \hat{\eta}^{(p)}_{j,t}$ where $G^{(p)}_{m,t}$ is an independent CHMM draw of the market factor and $\hat{\eta}^{(p)}_{j,t}$ is an empirical residual resampled with replacement.

The copula constructions use Sklar's theorem [11]: any joint CDF with continuous marginals factors as $H = C(F_1, \ldots, F_d)$. We probit-transform $u_{j,t} = \mathrm{rank}(g_{j,t})/(T+1)$, estimate Σ from pairwise Kendall's τ via $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [4], and for the Student-t copula select ν by profile MLE over $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Simulation reorders the ranks of each column of a per-asset CHMM draw to match the ranks of a copula sample [5], which preserves every asset's marginal exactly while injecting cross-asset dependence through rank coupling.

2.3 Evaluation

Every generator produces 1,000 independent simulated paths of length T (200 for Pipeline B, given the quadratic copula cost). Five complementary metrics score each path before aggregation: KS and Anderson-Darling pass rates at $\alpha = 0.05$; excess kurtosis; mean absolute error of the absolute-return autocorrelation function at lags 1-252 (ACF-MAE $= \frac{1}{L}\sum_{\tau=1}^{L} |\hat{\rho}^{\mathrm{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\mathrm{sim}}_{|G|}(\tau)|$); and Wasserstein-1 distance between pooled simulated and observed quantiles. Pipeline B additionally reports the Frobenius norm and off-diagonal MAE of the simulated Pearson correlation matrix relative to observed.

3 Empirical Study

3.1 Data

We use daily SPY prices from 2014-01-03 through 2024-01-03 as the in-sample (IS) window (T_{IS} = 2,516) and 2024-01-04 through 2026-04-17 as the out-of-sample (OoS) window (T_{OoS} = 572). Annualized excess log-returns are computed as $G_t = (1/\Delta t)\ln(P_t/P_{t-1}) - r_f$ with $\Delta t = 1/252$ and $r_f = 0$. The observed IS series is leptokurtic (excess kurtosis 7.68), has a linear ACF of returns that decays to noise within two lags, and an ACF of $|G_t|$ that remains statistically significant at lag 252. Figure 1 shows the four canonical stylized-fact diagnostics on the IS window.

3.2 Model selection: $K = 18$

A sweep over $K \in \{3, 6, 9, 12, 15, 18, 21\}$ under CHMM-N shows that $K = 18$ wins or ties on every distributional pass-rate metric (KS IS, KS OoS, AD IS, AD OoS, Wasserstein-1) and produces the highest simulated kurtosis (5.11 against observed 7.68). Margins between adjacent K values are under 1 percentage point on KS; the choice is supported by both the in-sample and out-of-sample pass-rate rankings.

3.3 SPY model comparison (Pipeline A)

Table 1 compares CHMM at $K = 18$ against the discrete-HMM baseline of [1] (NJ: no jumps; WJ: with Poisson jumps, $\epsilon = 0.01$, $\lambda = 3$) and four classical baselines. The CHMM variants achieve 94.7-95.2% IS KS pass rate and 80.4-84.0% OoS KS pass rate, in contrast to the 0% IS KS of both discrete variants (a direct consequence of returning bin-centroid values) and the 23.1% of GARCH(1,1). The i.i.d. bootstrap is the non-parametric ceiling at 99.6% IS KS but fails the temporal diagnostic (ACF-MAE 0.0627, the highest in the panel), confirming that good marginal fidelity alone does not certify a generator for time-series downstream use.

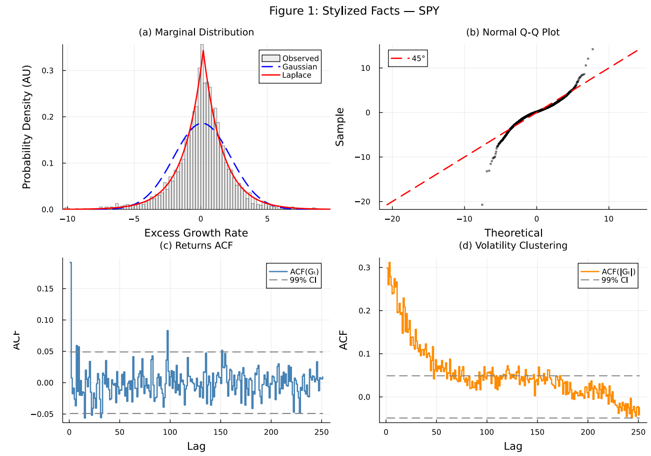

Figure 1: SPY stylized facts (IS, 2014-01-03 through 2024-01-03, T_{IS} = 2,516). Panel (a): marginal density with Gaussian and Laplace reference fits. Panel (b): normal Q-Q plot. Panel (c): ACF of G_t. Panel (d): ACF of $|G_t|$. Panels (a-b) evidence heavy tails; (c) evidences near-zero linear ACF; (d) evidences persistent volatility clustering.

Table 1: SPY model comparison, Pipeline A. 1,000 simulated paths, $\alpha = 0.05$, $K = 18$ for CHMM. IS: 2014-01-03 through 2024-01-03, T_{IS} = 2,516. OoS: 2024-01-04 through 2026-04-17, T_{OoS} = 572. Observed IS kurtosis 7.68. Column legend: KS = Kolmogorov-Smirnov pass rate, AD = Anderson-Darling pass rate, Kurt = mean simulated excess kurtosis, ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags, W_1 = Wasserstein-1 distance (IS).

	KS (%)		AD (%)				
Model	IS	OoS	IS	OoS	Kurt	ACF-MAE	W_1
Bootstrap	99.6	91.0	99.4	94.2	7.43	0.0627	0.065
Gaussian i.i.d.	0.0	2.0	0.0	0.0	-0.00	0.0627	0.411
Discrete NJ	0.0	38.1	0.0	65.9	0.64	0.0617	0.314
Discrete WJ	0.0	36.2	0.0	54.3	0.52	0.0618	0.326
GARCH(1,1)	23.1	58.3	8.5	55.5	6.97	0.0492	0.245
CHMM-N	94.7	84.0	86.5	76.5	5.02	0.0513	0.110
CHMM-t	94.7	83.4	89.9	79.0	17.34	0.0554	0.102
CHMM-L	95.2	80.4	93.2	79.9	6.76	0.0564	0.102

3.4 Six-ticker generalization (Pipeline A)

Table 2 generalizes the CHMM-N result from SPY to five additional tickers spanning distinct risk profiles: NVDA (high-volatility semiconductors), JNJ (defensive healthcare), JPM (financials), AAPL (large-cap tech), QQQ (tech-heavy index). Each ticker is fit independently with its own CHMM. IS KS pass rates are uniformly above 95%. OoS KS pass rates are above 90% on four of six tickers; NVDA (55.8%) and JPM (49.8%) show OoS degradation attributable to the

Table 2: Per-ticker CHMM-N generalization, Pipeline A. $K = 18$, 1,000 paths, $\alpha = 0.05$. Each ticker is fit independently; no cross-asset coupling. Complement: Table 3 evaluates cross-asset dependence on top of the same marginals.

	KS (%)		AD (%)		
Ticker	IS	OoS	IS	OoS	Kurt obs
SPY	95.5	81.4	88.9	76.5	7.7
NVDA	96.7	55.8	92.6	51.5	5.5
JNJ	99.4	93.9	98.8	92.1	9.0
JPM	97.8	49.8	94.4	48.6	7.6
AAPL	98.3	93.8	96.9	93.5	3.2
QQQ	95.5	92.3	87.8	95.0	3.4
Median	96.7	87.1	93.5	84.3	6.6

2024-2026 macroeconomic regime (the AI-driven semiconductor rally for NVDA, the interest-rate-repricing cycle for financials).

A quarterly walk-forward refit of the CHMM-N on the most-degraded ticker (JPM) lifts OoS KS from 49.0% to 64.3% and OoS AD from 48.6% to 63.6%, a 15-percentage-point recovery, confirming that the stationarity assumption of the fixed-IS fit is the dominant driver of the OoS gap on this ticker.

3.5 Cross-asset dependence (Pipeline B)

Table 3 evaluates three dependence constructions on top of the same CHMM-N marginals (Table 2): SIM with SPY as market factor, Gaussian copula, and Student-t copula with $\nu^* = 6$ selected by profile MLE. The two copula constructions preserve every per-asset marginal exactly (mean IS KS 97.5% / 96.2%); SIM's affine propagation degrades JPM (31.5% IS KS) and AAPL (64.0%) because the factor-resampled residuals re-enter each asset's marginal and thicken the tails beyond the asset's own distribution. On cross-asset correlation reproduction, the Student-t copula is the best (off-diagonal MAE 0.027, Frobenius 0.184), the Gaussian copula is second (0.031, 0.206), and the SIM trails (0.076, 0.527). The profile-MLE choice of $\nu^* = 6$ implies non-trivial joint tail dependence and is in the $4 \leq \nu \leq 10$ range typically reported for equity-index portfolios [3, 7].

4 Discussion and Conclusion

At a single moderate state count ($K = 18$) a continuous HMM trained by standard EM reproduces all three canonical stylized facts of equity returns without jump augmentation, without return quantization, and without additional tunable hyperparameters. Against the discrete-HMM baseline of [1] the IS KS pass-rate gap is 94.7 percentage points and is attributable to the move from bin-centroid emissions to continuous emissions: continuous emissions are a prerequisite for the simulated series to pass KS at all, and the elimination of Poisson-jump augmentation removes two hyper-parameters. The gap between Rydén et al. [9] (K = 2-3, failure to reproduce absolute-return ACF) and the present result (K = 18, ACF-MAE 0.0513) is a direct consequence of state resolution: sufficient diagonal dominance across a larger number of states approximates the slow ACF decay through a sum-of-exponentials, whereas K = 2-3 cannot.

Table 3: Pipeline B. Cross-asset dependence on fixed CHMM-N marginals. 200 simulated paths. Rows 1-6: per-asset IS KS (%), $\alpha = 0.05$ (sanity on the marginals after dependence injection). Rows 7-8: cross-asset correlation reproduction, $\|\hat{\Sigma}_{\mathbf{sim}} - \hat{\Sigma}_{\mathbf{obs}}\|_F$ and off-diagonal MAE, averaged over paths. Marginals identical across columns; only the dependence construction varies.

	SIM	Gaussian copula	Student-t copula ($\nu^* = 6$)
SPY	91.5	94.5	95.5
NVDA	71.5	96.0	93.0
JNJ	88.5	100.0	100.0
JPM	31.5	99.5	96.5
AAPL	64.0	99.5	99.0
QQQ	85.5	95.5	93.0
Mean KS	72.1	97.5	96.2
Median KS	78.5	97.8	96.0
Frobenius	0.527	0.206	**0.184**
Off-diag MAE	0.076	0.031	**0.027**

Pipeline B's ordering (Student-t copula over Gaussian copula over SIM) replicates the ordering reported by [1] with discrete marginals, suggesting the rank-reordering advantage is intrinsic to the construction and transfers across marginal families. Stationarity is the principal limitation: OoS pass rates drop on NVDA and JPM under the 2024-2026 regime, and a quarterly walk-forward refit recovers roughly 15 percentage points on JPM, supporting stationarity as the mechanism rather than mis-specified state count or emission family.

Scope. Option pricing and implied-volatility-surface calibration are out of scope. The six-asset universe is sufficient to establish the SIM-versus-copula ordering but does not address the scaling behavior of the copula construction to hundreds of assets, which would likely require vine or factor copulas [2].

Ethics and data provenance. All data used in this paper are daily OHLCV (open, high, low, close, volume) closing prices and volume-weighted average prices for six publicly-traded U.S. equities (SPY, NVDA, JNJ, JPM, AAPL, QQQ) retrieved from a commercial market-data vendor under a standard data-use license. No personally-identifying or privacy-sensitive information is used. The study presents no human-subject research.

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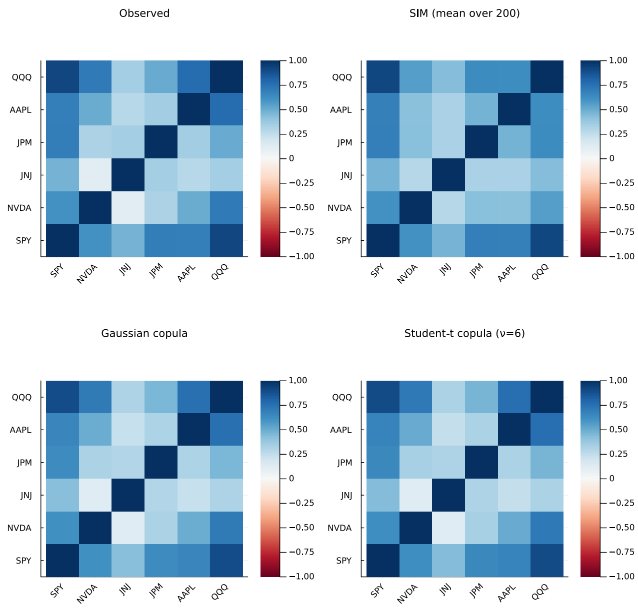

Figure 2: Pipeline B cross-asset correlation reproduction (companion to Table 3; IS window, $T_{\mathrm{IS}} = 2{,}516$, six tickers SPY, NVDA, JNJ, JPM, AAPL, QQQ). Panel (a): observed correlation matrix on IS returns. Panel (b): path-averaged correlation under the SIM with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM-N marginals. Panel (d): path-averaged correlation under the Student-t copula with $\nu^* = 6$. Each simulated panel averages 200 paths of length T_{IS}. Color scale in $[-1, 1]$, red-white-blue diverging; closer visual match to panel (a) is better. The SIM panel shows visibly larger residuals off the diagonal, matching the numerical ordering.

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